Databricks Certified Machine Learning Associate: 100 High-Probability Questions

Domain 1: Databricks Machine Learning (38%)

Q1. MLflow Tracking A data scientist needs parameters, metrics, and the model artifact logged automatically for a scikit-learn experiment. Which approach is best?

A. Manually write values to a CSV.

B. Use `mlflow.autolog()` or explicit `mlflow.log_param` and `log_metric` calls within a run.

C. Print values to the notebook output.

D. Save a screenshot of the experiment.

Q2. Model Registry A model must be promoted from development to production with a clear, auditable lifecycle. What is the recommended Databricks approach?

A. Email the pickle file to the production team.

B. Register the model in Unity Catalog with versions and an alias such as `@champion`.

C. Rename the artifact folder.

D. Copy the model file to a shared DBFS path.

Q3. Feature Store A team wants to avoid training-serving skew. Which Databricks capability is designed for this?

A. A shared Excel spreadsheet of feature definitions.

B. Unity Catalog Feature Store with point-in-time joins.

C. Recomputing features separately in the serving code.

D. Using only raw columns at inference.

Q4. AutoML A data scientist wants a quick baseline on a labeled tabular dataset. Which Databricks feature is most appropriate?

A. Databricks AutoML.

B. Building a custom neural network from scratch.

C. Deploying an endpoint before training.

D. Creating an online feature table only.

Q5. ML Runtime What is a primary advantage of using a Databricks Runtime for Machine Learning cluster?

A. It includes common ML libraries and integrations preinstalled.

B. It eliminates all package conflicts automatically.

C. It removes the need for cluster permissions.

D. It allows skipping feature engineering entirely.

Q6. Unity Catalog Registry What is a key benefit of registering models in Unity Catalog rather than the workspace registry?

A. Models become accessible only to the original author.

B. Models are governed, discoverable, and promotable across workspaces at the account level.

C. Models no longer need a version number.

D. Models are automatically deployed to serving endpoints.

Q7. MLflow Search A data scientist has many MLflow runs. How can they programmatically find the best run by a metric?

A. Use `mlflow.search_runs()` and sort the resulting DataFrame.

B. Manually click through each run in the UI.

C. Export all runs to a CSV and use Excel.

D. Delete runs until only one remains.

Q8. Model Aliases What replaced the older model stages (Staging, Production, Archived) in Unity Catalog Model Registry?

A. Numeric version numbers only.

B. Aliases such as `champion` and `challenger`.

C. Tags only.

D. File system paths.

Q9. Online vs Offline Features What is the primary difference between online and offline feature tables?

A. Online tables are for training; offline tables are for serving.

B. Online tables serve low-latency inference; offline tables are for batch training and analysis.

C. Online tables cannot be created in Unity Catalog.

D. Offline tables do not support point-in-time joins.

Q10. Feature Engineering Client Which class is used to create and manage feature tables in Unity Catalog?

A. `FeatureEngineeringClient`.

B. `SparkSession`.

C. `MLflowClient`.

D. `Pipeline`.

Q11. AutoML Output What useful artifact does Databricks AutoML generate that allows a team to review and customize the winning approach?

A. A screenshot of the leaderboard only.

B. Generated notebooks for data exploration and the best trial.

C. A compiled binary with no source code.

D. A cluster event log.

Q12. MLOps Best Practice Which setup best reflects a sound MLOps strategy?

A. One shared workspace where notebooks are edited directly in production.

B. Separate development, staging, and production environments with versioned code and tracked artifacts.

C. Manual model file copies between users.

D. Training new models without keeping prior run metadata.

Q13. Reproducibility Which practice most directly improves reproducibility in a Databricks ML workflow?

A. Keeping training code in source control and logging run parameters and artifacts with MLflow.

B. Relying on notebook revision history alone.

C. Letting each data scientist rename columns locally.

D. Saving only final predictions.

Q14. Model Promotion Strategy In which scenario is promoting code preferred over promoting models?

A. When the model is a simple scikit-learn model.

B. When the training pipeline is complex and the model artifact is small.

C. When the model artifact is large and training is expensive.

D. When the team uses only AutoML.

Q15. MLflow UI Where in the MLflow UI can a user find the notebook that was run to create an experiment run?

A. The "Source" link for the run.

B. The "Artifacts" tab only.

C. The cluster event log.

D. The Databricks home page.

Q16. Feature Store Creation Which method creates a feature table in Unity Catalog?

A. `fe.create_table()`.

B. `spark.sql("CREATE TABLE")`.

C. `mlflow.log_artifact()`.

D. `dbutils.fs.mkdirs()`.

Q17. Point-in-Time Lookup Why is point-in-time lookup important when creating training sets from a feature store?

A. It ensures training does not use feature values that were unavailable at prediction time.

B. It speeds up the query.

C. It automatically imputes missing values.

D. It converts categorical features to numeric.

Q18. Model Tags How can a tag be set on a registered model in Unity Catalog?

A. `mlflow.set_model_tag()`.

B. `spark.conf.set()`.

C. `dbutils.notebook.exit()`.

D. `display()`.

Q19. MLflow Run Identification A data scientist wants to identify the best run using the MLflow Client API. Which method is used?

A. `client.search_runs()`.

B. `client.list_experiments()`.

C. `client.get_run()`.

D. `client.delete_run()`.

Q20. Champion Promotion How is a challenger model promoted to champion using aliases?

A. Delete the old model and upload the new one.

B. Move the `@champion` alias to the new model version.

C. Rename the model.

D. Create a new experiment.

Domain 2: Model Development (31%)

Q21. Regression Metrics A regression model has RMSE = 25, MAE = 20, and $R^2 = 0.85$. Which statements are correct? (Select two.)

A. The model captures 85% of the variance in the target.

B. The RMSE being higher than MAE suggests larger errors exist.

C. The model is underfitting.

D. MAE is 25.

Q22. Imbalanced Classification A binary classifier reports 98% accuracy on data with 2% positives. What should be done?

A. Ship the model immediately.

B. Evaluate precision, recall, F1, and PR-AUC; consider class weights or resampling on training data only.

C. Increase training epochs.

D. Remove the positive class.

Q23. One-Hot Encoding When is one-hot encoding least appropriate?

A. When the categorical feature has low cardinality.

B. When the categorical feature has extremely high cardinality.

C. When using linear models.

D. When the feature is a string.

Q24. Imputation For a feature with a highly skewed distribution, which imputation strategy is generally preferred?

A. Mean.

B. Median.

C. Mode.

D. Zero.

Q25. Log Transformation When is a log transformation most appropriate?

A. When the feature is normally distributed.

B. When the feature is right-skewed and spans several orders of magnitude.

C. When the feature is categorical.

D. When the target is binary.

Q26. Estimator vs Transformer In Spark ML, what is the key difference between an Estimator and a Transformer?

A. An Estimator has a `fit()` method; a Transformer has a `transform()` method.

B. An Estimator only transforms data; a Transformer fits models.

C. They are identical.

D. A Transformer cannot be used in a Pipeline.

Q27. Spark ML Pipeline Why should preprocessing and model training be wrapped in a Spark ML Pipeline?

A. To ensure the same preprocessing is applied at inference as during training.

B. To reduce the number of columns.

C. To avoid using MLflow.

D. To eliminate the need for a test set.

Q28. Hyperopt Parallelization Which Hyperopt component distributes trials across a Spark cluster?

A. `SparkTrials`.

B. `Trials`.

C. `fmin`.

D. `hp.uniform`.

Q29. Hyperopt Search Algorithm Which search algorithm does Hyperopt use by default via `tpe.suggest`?

A. Grid search.

B. Random search.

C. Tree-structured Parzen Estimator (Bayesian optimization).

D. Exhaustive search.

Q30. Cross-Validation vs Train-Validation Split What is the main trade-off between cross-validation and a single train-validation split?

A. Cross-validation is faster.

B. Cross-validation gives a more stable estimate but is more computationally expensive.

C. Train-validation split always gives higher accuracy.

D. They are identical in cost.

Q31. Classification Metrics Which metric is most appropriate for a highly imbalanced binary classification task?

A. Accuracy.

B. Area under the Precision-Recall curve (PR-AUC).

C. Mean Absolute Error.

D. R^2.

Q32. Random Split Which Spark operation randomly splits a DataFrame into training and test sets?

A. `DataFrame.randomSplit()`.

B. `TrainValidationSplit`.

C. `CrossValidator`.

D. `DataFrame.where()`.

Q33. StringIndexer Before one-hot encoding a string categorical column in Spark ML, which stage is typically required?

A. `StringIndexer`.

B. `VectorAssembler`.

C. `StandardScaler`.

D. `Imputer`.

Q34. VectorAssembler What is the purpose of `VectorAssembler` in Spark ML?

A. To combine multiple feature columns into a single vector column.

B. To impute missing values.

C. To encode categorical variables.

D. To split the data.

Q35. Feature Scaling Which scaler transforms features to have zero mean and unit variance?

A. `MinMaxScaler`.

B. `StandardScaler`.

C. `MaxAbsScaler`.

D. `Normalizer`.

Q36. Data Leakage A data scientist standardizes features before splitting into train and test. What is the risk?

A. The test set statistics leak into training, giving an overly optimistic evaluation.

B. The model will underfit.

C. The features will become categorical.

D. Nothing; this is best practice.

Q37. Pandas UDF What is a benefit of vectorized pandas UDFs over standard PySpark UDFs?

A. They process data in batches rather than row by row, improving performance.

B. They work only on single-node DataFrames.

C. They require no Python.

D. They automatically handle missing values.

Q38. Target Encoding Target encoding is most useful for which situation?

A. Low-cardinality categorical features.

B. High-cardinality categorical features.

C. Continuous features with outliers.

D. Binary targets only.

Q39. Overfitting A model performs very well on training data but poorly on test data. What is this called?

A. Underfitting.

B. Overfitting.

C. Data leakage.

D. Regularization.

Q40. Bias-Variance High variance in a model typically corresponds to which problem?

A. Overfitting.

B. Underfitting.

C. Data imbalance.

D. Feature scaling.

Q41. Median vs Mean Imputation When is median imputation preferred over mean imputation?

A. When the feature is symmetric.

B. When the feature has outliers or is skewed.

C. When the feature is categorical.

D. When there are no missing values.

Q42. Class Weights How can class imbalance be mitigated without resampling?

A. Use class weights in the model training.

B. Remove all minority samples.

C. Duplicate the majority class.

D. Ignore the problem.

Q43. Evaluation Metric Selection For a regression task where large errors are especially costly, which metric is most appropriate?

A. RMSE.

B. MAE.

C. Accuracy.

D. F1.

Q44. Spark ML Model Persistence How is a fitted Spark ML model saved?

A. `model.write().overwrite().save(path)`.

B. `mlflow.log_artifact()`.

C. `spark.sql("SAVE")`.

D. `dbutils.fs.put()`.

Domain 3: ML Workflows / Data Processing (19%)

Q45. Summary Statistics Which command computes summary statistics on a Spark DataFrame?

A. `df.summary()`.

B. `df.show()`.

C. `df.printSchema()`.

D. `df.count()`.

Q46. Outlier Removal How can outliers be removed based on standard deviation?

A. Filter values beyond a chosen number of standard deviations from the mean.

B. Remove all values above the median.

C. Remove the top 50% of values.

D. Use `dropDuplicates()`.

Q47. IQR Outlier Removal What is the IQR method for outlier detection?

A. Remove values below Q1 or above Q3.

B. Remove values below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$.

C. Remove values beyond the mean.

D. Remove values with z-score > 1.

Q48. Pandas API on Spark A data scientist has a pandas notebook and needs to scale to big data with minimal refactoring. What is the best approach?

A. Rewrite everything in Scala.

B. Refactor to use the pandas API on Spark.

C. Use only Spark SQL.

D. Sample the data to fit in memory.

Q49. Visualization Which method is appropriate for comparing two continuous features?

A. A scatter plot.

B. A bar chart of counts.

C. A pie chart.

D. A box plot of a single feature.

Q50. Categorical Comparison Which visualization is appropriate for comparing two categorical features?

A. A grouped bar chart or contingency table heatmap.

B. A scatter plot.

C. A line chart.

D. A histogram of a continuous variable.

Q51. Missing Value Strategy For a categorical feature, which imputation strategy is most appropriate?

A. Mean.

B. Median.

C. Mode.

D. Linear interpolation.

Q52. Distribution Comparison Which method compares two continuous distributions?

A. Overlaid histograms or density plots.

B. A single pie chart.

C. A bar chart of category counts.

D. A confusion matrix.

Q53. Feature Store Training Set Which method creates a training set from a feature table with point-in-time lookups?

A. `fe.create_training_set()`.

B. `spark.read.table()`.

C. `mlflow.log_model()`.

D. `Pipeline.fit()`.

Q54. Online Table Serving What is the primary purpose of an online feature table?

A. Low-latency feature lookup for real-time inference.

B. Batch training.

C. Long-term storage.

D. Data visualization.

Q55. Spark DataFrame Random Split Which method splits a DataFrame into train and test sets with a seed?

A. `df.randomSplit([0.8, 0.2], seed=42)`.

B. `df.split(0.8)`.

C. `TrainValidationSplit`.

D. `CrossValidator`.

Q56. CrossValidator What does `CrossValidator` perform in Spark ML?

A. k-fold cross-validation for model selection and hyperparameter tuning.

B. A single train-test split.

C. Feature scaling.

D. Model deployment.

Q57. TrainValidationSplit What does `TrainValidationSplit` do?

A. Splits data once into training and validation sets for tuning.

B. Performs k-fold cross-validation.

C. Deploys a model.

D. Encodes categorical variables.

Q58. Log Scale When should a log transformation be applied to a feature?

A. When the feature is right-skewed or spans orders of magnitude.

B. When the feature is binary.

C. When the feature has no variance.

D. When the feature is already normally distributed.

Q59. Outlier Impact Why can outliers be problematic for mean imputation?

A. The mean is sensitive to outliers, so the imputed value may be biased.

B. Outliers make the mean undefined.

C. The median is always worse.

D. Outliers improve the mean.

Q60. Data Summary Which command provides descriptive statistics including count, mean, std, min, and max?

A. `df.describe()`.

B. `df.show()`.

C. `df.columns`.

D. `df.dtypes`.

Q61. Categorical Encoding What is the main drawback of one-hot encoding for very high-cardinality features?

A. It creates a very large number of sparse columns.

B. It loses information.

C. It cannot be used with Spark ML.

D. It requires the target variable.

Q62. Mode Imputation When is mode imputation most appropriate?

A. For continuous features with outliers.

B. For categorical features with missing values.

C. For binary targets.

D. For normally distributed continuous features.

Q63. Pandas API Benefit What is a key benefit of pandas API on Spark?

A. It allows pandas users to scale to big data with minimal code changes.

B. It eliminates the need for Spark.

C. It only works on single-node clusters.

D. It replaces MLflow.

Domain 4: Model Deployment (12%)

Q64. Serving Endpoint What is the purpose of a Databricks serving endpoint?

A. To provide a REST API for real-time model inference.

B. To store training data.

C. To run SQL queries.

D. To manage cluster permissions.

Q65. Scale to Zero An ML engineer wants a serving endpoint active only during working hours to minimize cost. What should be enabled?

A. Scale to Zero.

B. Autoscaling only.

C. A larger cluster.

D. Model versioning.

Q66. Batch vs Real-Time What is the difference between batch inference and real-time inference?

A. Batch processes large volumes on a schedule; real-time serves individual requests with low latency.

B. Batch is always faster.

C. Real-time cannot use MLflow models.

D. They are identical.

Q67. Model Deployment Registration Before deploying a model to a serving endpoint, what step is typically required?

A. Register the model in Unity Catalog.

B. Delete all prior runs.

C. Convert the model to CSV.

D. Stop the cluster.

Q68. Traffic Splitting What is the purpose of traffic splitting between serving endpoints?

A. To route a percentage of traffic to different model versions for A/B testing.

B. To increase training speed.

C. To reduce storage costs.

D. To impute missing values.

Q69. Streaming Inference Which Databricks capability supports streaming inference?

A. Delta Live Tables with model serving.

B. Only batch inference.

C. Only real-time endpoints.

D. Only AutoML.

Q70. Model Serving Scaling What does autoscaling for serving endpoints do?

A. Adjusts the number of serving instances based on traffic.

B. Automatically trains new models.

C. Changes the model version.

D. Deletes old experiments.

Q71. Deployment Consistency Why must feature engineering be identical between training and serving?

A. To avoid training-serving skew and ensure consistent predictions.

B. To reduce storage costs.

C. To speed up cluster startup.

D. To avoid using Unity Catalog.

Q72. Model Version Deployment How can a specific model version be served from a Unity Catalog registry?

A. Reference the model by its version or alias in the serving endpoint configuration.

B. Export the model to a CSV file.

C. Rename the model.

D. Use only the latest version automatically.

Q73. Deployment Monitoring What is a key aspect of monitoring a deployed model?

A. Tracking prediction drift and data drift over time.

B. Only monitoring cluster CPU usage.

C. Ignoring performance after deployment.

D. Only monitoring training accuracy.

Additional High-Frequency Recall Items

Q74. MLflow Autologging What does `mlflow.autolog()` do?

A. Automatically logs parameters, metrics, and models for supported libraries.

B. Automatically deploys a model.

C. Automatically creates a cluster.

D. Automatically deletes old runs.

Q75. Artifact Logging How is a file artifact logged to an MLflow run?

A. `mlflow.log_artifact(local_path)`.

B. `mlflow.log_metric()`.

C. `mlflow.log_param()`.

D. `mlflow.start_run()`.

Q76. Parameter Logging Which function logs a hyperparameter to MLflow?

A. `mlflow.log_param()`.

B. `mlflow.log_metric()`.

C. `mlflow.log_artifact()`.

D. `mlflow.register_model()`.

Q77. Metric Logging Which function logs a numeric metric to MLflow?

A. `mlflow.log_metric()`.

B. `mlflow.log_param()`.

C. `mlflow.log_artifact()`.

D. `mlflow.set_tag()`.

Q78. Start Run Which function starts a new MLflow run?

A. `mlflow.start_run()`.

B. `mlflow.end_run()`.

C. `mlflow.search_runs()`.

D. `mlflow.log_model()`.

Q79. Register Model Which function registers a model in Unity Catalog?

A. `mlflow.register_model()`.

B. `mlflow.log_model()`.

C. `mlflow.save_model()`.

D. `mlflow.export_model()`.

Q80. Spark ML Evaluator Which class evaluates a binary classifier in Spark ML?

A. `BinaryClassificationEvaluator`.

B. `RegressionEvaluator`.

C. `MulticlassClassificationEvaluator` only.

D. `ClusteringEvaluator`.

Q81. Regression Evaluator Which metric is available in `RegressionEvaluator`?

A. RMSE.

B. Precision.

C. Recall.

D. F1.

Q82. Multiclass Evaluator Which metric is available in `MulticlassClassificationEvaluator`?

A. F1.

B. RMSE.

C. MAE.

D. R^2.

Q83. Feature Importance How can feature importance be accessed for a tree-based Spark ML model?

A. `model.featureImportances`.

B. `model.coefficients`.

C. `model.intercept`.

D. `model.summary`.

Q84. Linear Model Coefficients How are coefficients accessed for a linear regression model in Spark ML?

A. `model.coefficients`.

B. `model.featureImportances`.

C. `model.intercept` only.

D. `model.summary` only.

Q85. Intercept How is the intercept of a linear model accessed?

A. `model.intercept`.

B. `model.coefficients`.

C. `model.featureImportances`.

D. `model.summary`.

Q86. StringIndexer Handle Invalid What happens by default when `StringIndexer` encounters an unseen label at transform time?

A. It throws an error unless `handleInvalid` is set.

B. It silently ignores the row.

C. It maps to index 0.

D. It maps to the most frequent label.

Q87. OneHotEncoder Input What type of column does `OneHotEncoder` expect as input in Spark ML?

A. A numeric index column, typically from `StringIndexer`.

B. A raw string column.

C. A continuous feature.

D. A vector column.

Q88. VectorAssembler Input What types of columns can `VectorAssembler` accept?

A. Numeric, boolean, and vector columns.

B. Only string columns.

C. Only categorical columns.

D. Only binary columns.

Q89. Spark ML PipelineModel What does fitting a `Pipeline` produce?

A. A `PipelineModel` that can be used for transformation and prediction.

B. A raw DataFrame.

C. A cluster.

D. An MLflow experiment.

Q90. Model Persistence Format What format does Spark ML use to save models?

A. A directory containing metadata and Parquet files.

B. A single CSV file.

C. A pickle file only.

D. A JSON file only.

Q91. MLflow Model Flavor What is an MLflow model flavor?

A. A convention for saving and loading models from different libraries (e.g., sklearn, spark).

B. A type of cluster.

C. A visualization tool.

D. A data source.

Q92. Unity Catalog Three-Level Namespace What is the three-level namespace for a model in Unity Catalog?

A. `catalog.schema.model`.

B. `workspace.model.version`.

C. `database.table.column`.

D. `user.model.run`.

Q93. Model Signature What is the purpose of a model signature when logging with MLflow?

A. It defines the expected input and output schema for the model.

B. It encrypts the model.

C. It compresses the model.

D. It deletes the model after logging.